Physics 336
Winter 2026
Final Exam

Name________________________________

Fill In the Blank______________________

1. (20 points) An experiment measures two quantities (suppose we call them x and y). Systematic errors are well controlled so that there are present only random instrumentation errors. The quantity x was measured for 10.0 s, with the result $x = 2.9 \pm 0.4$ cm. the quantity y was measured for 15.0 s, with the result $y = 4.65 \pm 0.08$ cm. Now suppose that the experimental conditions are the same, but that x is measured for 40.0 s and y is measured for 240. s. For this case, compute

$$z = 3\cos\left(\frac{\pi x}{y}\right)$$

with its uncertainty.

2. Consider several cubic samples (of known masses, m) of a particular metal alloy. The lengths, s of one side of these cubes are measured:

$$m = \{4.50, 6.00, 7.50, 9.00, 12.00, 15.00, 18.00\} \text{ g}$$
$$s = \{1.07, 1.14, 1.24, 1.32, 1.41, 1.57, 1.65\} \text{ cm}$$

   (a) (7 points) Determine an expected relationship for $s(m)$.
   (b) (8 points) Fit the data using a non-linear fit.
   (c) (8 points) Create a *linear* fit.
   (d) (7 points) Use your results to calculate the density of this alloy.